

Jalen Edusei

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EDUCATION

University of Georgia, Morehead Honors College, Athens, GA
Bachelor of Science, Computer Systems Engineering

May 2026
GPA: 3.61

Relevant Coursework: Data Structures, Systems Programming, Software Development, Discrete Mathematics, Advanced Digital Design, Design of Digital Systems, Embedded Systems I & II, Sensors & Transducers, Linear Systems, Human-Computer Interaction, Microfabrication, Probability & Statistics for Engineering.

RESEARCH EXPERIENCE

Joyner Research Laboratory, *Research Assistant*, Athens, GA

September 2022 – May 2023

- Conducted 50+ experiments including ELISAs and DNA/RNA extractions supporting malaria bioinformatics projects.
- Analyzed and documented data contributing to 2 peer-reviewed manuscripts.
- Designed and launched a new laboratory website, increasing research visibility and engagement by over 500 monthly visitors.

Small Satellite Research Laboratory, *Flight Software Developer (MEMESat-1)*, Athens, GA, March 2024 – December 2024

- Developed and debugged embedded flight software in C/C++ for the MEMESat-1 CubeSat on STM32 microcontrollers, bringing up and troubleshooting UART communication between onboard subsystems.
- Authored unit and integration tests to verify software behavior and harden the flight system against edge cases and failure conditions.
- Reviewed teammates' code and documentation, providing feedback that improved consistency, readability, and overall quality across the flight software codebase.

University of Georgia, *Capstone Researcher (AnimalDot)*, Athens, GA

August 2025 – May 2026

- Designed a contactless smart sensing bed to monitor animal heart rate and respiration using geophone-based vibration sensing, prioritizing minimal animal disturbance and continuous passive data collection.
- Developed a modular sensing and processing pipeline integrating geophones, load cells, and temperature sensors with analog signal conditioning, filtering, and feature extraction to separate physiological signals from environmental noise.
- Architected an embedded-to-mobile system that transmits processed data to a companion mobile application for real-time visualization, trend tracking, and caregiver-facing health insights.

TEACHING & MENTORING EXPERIENCE

Office of Engagement, Leadership, and Service, *ELS Peer Leader*, Athens, GA

January 2024 – May 2024

- Selected for leadership coaching program supporting first- and second-year student engagement in UGA involvement pathways.
- Conducted 1:1 mentorship sessions, connecting students to leadership opportunities, clubs, and resources across campus.
- Facilitated interactive workshops that improved student understanding of values-based leadership and self-discovery.

UGA STEM Academic Success Program, *Student Advisor*, Athens, GA

August 2023 – May 2025

- Planned and executed 8+ workshops each semester supporting academic success for 30+ STEM students.
- Assisted in strategic programming and mentoring initiatives contributing to increased retention and engagement.
- Developed and maintained the organization's digital and social media strategy, increasing online engagement by 50%.

University of Georgia Housing, *Resident Assistant*, Athens, GA

August 2023 – May 2025

- Cultivated a supportive community for 45 residents by organizing 10+ educational and social events each semester, boosting resident engagement by 30%.
- Mediated and resolved 30+ conflicts and safety concerns, maintaining a secure environment.

- Partnered with housing staff to implement programming focused on academic success and mental health awareness.

TECHNICAL & ENGINEERING PROJECTS

Personal Portfolio Website

Athens, GA

Design Engineering & Full Stack Development

January 2026 – present

- Architected an immersive 3D workstation experience using React Three Fiber and Framer Motion, implementing high-fidelity spatial transitions and intentional motion patterns between 2D and 3D states.
- Engineered a robust component architecture and custom theme engine with Tailwind CSS and Zustand, ensuring synchronized state management across complex 3D overlays and traditional web interfaces.
- Optimized 3D asset delivery and interaction performance for a Progressive Web App (PWA) environment, maintaining a consistent 60 FPS experience and mobile-first responsiveness across all major browsers.
- Designed a terminal-inspired navigation system and modular asset pipeline, establishing end-to-end technical ownership from Figma-based prototyping to production-ready deployment on Vercel.

Musical Artist Website and Storefront

Athens, GA

Freelance Full Stack Development

May 2026 – present

- Built a production music-artist site and e-commerce storefront in Next.js 16 (App Router) and React 19 / TypeScript, integrating the Shopify Storefront API as a headless commerce backend with cart state persisted in an httpOnly cookie and all cart mutations routed through a hardened server-side API boundary.
- Modeled all editorial content (releases, tracks, shows, videos, press, and lookbooks) in Sanity CMS with an embedded Studio, and wired on-demand Incremental Static Regeneration via HMAC-signed Sanity webhooks to revalidate only the paths affected by each content change.
- Implemented a Klaviyo-backed email capture flow with Zod schema validation and Upstash Redis per-IP rate limiting (429 with **Retry-After**), organizing the marketing and store experiences into separate route groups under a shared Tailwind CSS v4 design system.
- Instrumented the platform with Sentry error monitoring and Vercel Analytics and Speed Insights, delivering a mobile-first responsive build deployed end-to-end on Vercel.

ATLAS Job Search Platform

Athens, GA

AI Application Development & Full Stack Engineering

April 2026

- Built a full-stack React job-search operating system featuring a Kanban-style pipeline visualization and full CRUD application tracking, centralizing every stage of the search from outreach to offer in a single dashboard.
- Integrated the Anthropic Claude API to power a suite of AI tools including resume tailoring, cover letter generation, company research, cold outreach drafting, interview preparation, mock interviews, and offer comparison.
- Engineered an analytics layer with interactive charts surfacing application velocity, response rates, and pipeline health, enabling data-informed decisions across the search.

Übersicht Desktop Widget Suite

Athens, GA

Frontend Engineering & Creative Coding

May 2026

- Built a suite of 12 macOS desktop widgets for Übersicht in React/JSX, unifying them under a single self-contained design system of color tokens, typography, a frosted-glass card shell, drag/resize handles, and last-known-good caching for consistent look and offline resilience.
- Engineered graphics-heavy widgets including a full-screen WebGL fragment-shader live wallpaper, an auto-rotating 3D PBR model viewer with daily rotation and offline fallback, and a dot-matrix globe rendering great-circle arcs and pins over visited cities.
- Integrated external data sources through shell-backed commands, surfacing Apple Music now-playing and most-played albums, NASA's Astronomy Picture of the Day with inline video, and Claude API-generated daily prompts.
- Packaged each widget as an independent GitHub repository with automated shell install scripts and distributable bundles, owning the project end-to-end from design system to documentation and release.

PDMS Microfluidic Mixer

Athens, GA

for Microfabrication

January 2026 – May 2026

- Designed and fabricated passive microfluidic mixing devices in PDMS to overcome diffusion-limited mixing in laminar, low-Reynolds-number flow, comparing Y-junction, sharp-bend, and smooth-bend serpentine channel geometries to maximize fluid interface generation and in-channel residence time.
- Produced photoresist molds via photolithography (1000 RPM spin coating, 95 °C soft bake, mask-aligner UV

exposure, post-exposure bake, and development) and cast PDMS devices using trichlorosilane surface modification, plasma bonding to glass, and thermal curing.

- Characterized fabricated devices on a Chemyx syringe pump across input flow rates from 0.001 to 0.1 mL/min, capturing microscope imagery at junctions and bends to evaluate mixing performance and diagnose dust-defect and channel-leakage failure modes.
- Collaborated on a four-person team to document results and propose future work, including quantitative mixing metrics, varied channel heights, and fluids of differing viscosities and Reynolds numbers.

PrimeForge FPGA Prime Number Detector

for Advanced Digital Systems, CSEE 4280

Athens, GA

January 2026 – May 2026

- Architected a Verilog-based prime number detection system on a Xilinx FPGA, unifying a VGA output controller, SD card interface, DDR2 memory arbiter, and parallel compute engines under a single top-level module.
- Implemented dual prime-finding cores: a Sieve of Eratosthenes engine and a trial-division engine, with a configurable 64-bit upper bound, enabling runtime selection between throughput-optimized and resource-efficient computation paths.
- Designed a 640x480 @ 60Hz VGA pipeline with a sprite engine, 8x16 font ROM text renderer, and DDR2-backed frame buffering to display the last 20 primes, runtime, and accuracy against a reference set loaded from SD card over SPI.
- Built a multi-input user interface combining PMOD joystick, keypad, debounced buttons, and an FSM-driven menu, and verified the design with module-level testbenches for the sieve engine, trial divider, DDR2 arbiter, SD SPI controller, sprite engine, and VGA timing.

PARMCO Bluetooth Motor Control System

for Embedded Systems II, ECSE 4235

Athens, GA

January 2026 – May 2026

- Built a BLE GATT server in C on a Raspberry Pi 4 that advertises a custom service, parses UTF-8 motor control commands, and streams RPM telemetry notifications to a SwiftUI iOS client over Core Bluetooth.
- Integrated ARMv8 assembly GPIO primitives (via direct register mapping) with the pigpio DMA-based hardware PWM library to drive an L293D motor driver, supporting forward/reverse direction, variable speed, and start/stop commands from the phone.
- Implemented closed-loop control modes (Maintain and Match) that sample an IR tachometer, compute actual RPM, and adjust PWM duty cycle in real time to track an RPM setpoint or a reference tach input.
- Packaged the server as a systemd service for boot auto-start and designed the SwiftUI client with manual and automatic control modes, delivering a demo-ready phone-to-motor system with bidirectional BLE communication.

AnimalDot Smart Bed

for Capstone Design

Athens, GA

August 2025 – May 2026

- Engineered ESP32 firmware in C++ for a contactless animal vital-signs monitoring bed, orchestrating a geophone (heart rate / respiration), DHT22 temperature sensor, and FX29 load cells through a modular sensor manager with NVS-persisted calibration.
- Implemented a real-time DSP pipeline (band-pass filtering, peak-count detection, and exponential smoothing) to extract cardiac and respiratory signals from raw geophone vibration data while suppressing environmental noise.
- Architected concurrent BLE, WiFi, and MQTT connectivity stacks, including NimBLE GATT services with the standard 0x180D Heart Rate profile, STA auto-connect with captive-portal AP fallback, and BedDot-compatible binary MQTT payloads for cloud telemetry.
- Delivered an end-to-end embedded-to-mobile system with a React Native companion app, Node.js backend, and Next.js caregiver dashboard, providing live vital sign visualization, trend tracking, and remote configuration of the bed.

Live Election Platform

for a campus engineering student organization

Athens, GA

March 2026 – April 2026

- Designed and shipped a presenter-paced live election system in Next.js 14 (App Router) and Tailwind CSS, supporting 60–80 concurrent voters across 14 parliamentary roles and 39 candidates with a finite-state election machine (Waiting → Voting → Locked → Results) and runoff/floor-nomination support.
- Architected the realtime layer on Supabase Postgres with WebSocket Broadcast for sub-100ms ballot delivery, server-synced countdowns via absolute UTC `poll_expires_at` timestamps, and 0–400ms randomized vote jitter to mitigate stampeding-herd load at submission.
- Engineered a defense-in-depth anti-fraud stack combining composite SHA-256 browser fingerprinting (Canvas + AudioContext + system entropy), a database-level `UNIQUE(role_id, device_hash)` constraint with silent SQLSTATE 23505 handling, and Row-Level Security policies to prevent duplicate voting and result tampering.
- Built an admin dashboard with progressive disclosure (pre-poll checklist, live telemetry, results with tie detection)

and a WCAG AA voter UI featuring 48dp touch targets, `role="timer"` live regions, Web Worker heartbeats, and REST fail-safe reconnection on `visibilitychange`.

Kitchen Chaos VR

for Virtual Reality, CSCI 6830

Athens, GA

October 2025 – December 2025

- Built an Overcooked-style multiplayer VR cooking game for Meta Quest 3 in Unity, implementing physics-driven interactions, locomotion, and object handling across a modular C# gameplay architecture.
- Integrated VelNet networking to synchronize player avatars, spawning, and round state across clients, enabling reliable session flow from bootstrap to gameplay with deterministic event handling and scene management.
- Developed a recipe and scoring pipeline using ScriptableObjects, trigger-based ingredient tracking, and interpretable score breakdowns, then integrated an AI dish judge via REST API calls with strict JSON outputs and text-to-speech narration.

BreakBuddy

for Human-Computer Interaction, CSCI 4800

Athens, GA

August 2025 – December 2025

- Designed BreakBuddy, a guilt-free stress management application for educators that converts 2 to 5 minute pauses into guided micro-break sessions with social accountability and low cognitive load.
- Conducted user research through semi-structured interviews, affinity mapping, POV and HMW framing, and assumption mapping, then validated concepts via experience prototyping and iterative heuristic evaluation.
- Built and tested paper and high-fidelity prototypes featuring a robust activity timer, defensive design error states, and a Reports dashboard with streak tracking and data visualization.

Virtual Reality Portfolio 2

for Virtual Reality, CSCI 6830

Athens, GA

October 2025 – November 2025

- Developed two advanced XR experiences for the Meta Quest 3, including a VR Mini Museum and an MR Room, featuring spatial audio storytelling, physics-based interactions, and hand-tracked UI.
- Implemented mixed reality features such as depth-based occlusion, passthrough visualization, spatial anchors, distance grabbing, and real-time object spawning across a modular system of more than 20 C# scripts.
- Built an AI-driven NPC assistant using Wit.ai for natural language understanding, speech synthesis, and lip-synchronized responses that enable voice-controlled panels.

Smart Plant Watering Assistant

for Sensors and Transducers, ELEE 4230

Athens, GA

August 2025 – November 2025

- Built an automated plant monitoring system using a Raspberry Pi Pico 2W connected to a soil moisture sensor, thermistor, and LDR to analyze environmental conditions in real time.
- Integrated analog signal conditioning, operational amplifier stages, and Kalman filtering to stabilize noisy sensor data and enhance measurement precision.
- Implemented control logic for a transistor-driven water pump with predictive thresholding to maintain optimal soil moisture, demonstrating reliable closed-loop system performance.

Virtual Reality Portfolio 1

for Virtual Reality, CSCI 6830

Athens, GA

August 2025 – October 2025

- Constructed a four-part Unity portfolio demonstrating key VR design principles including transformation, physics-based motion, immersion, and user interaction with spatial environments.
- Authored over 15 C# scripts integrating real-time lighting, spatial audio, collision detection, and smooth locomotion, translating theoretical VR concepts into practical, interactive learning modules.

Audio Tracking Car

for ECSE Design Methodology, ECSE 2920

Athens, GA

January 2025 – April 2025

- Engineered a Python-based control system on Raspberry Pi 4 that autonomously navigated towards specific audio frequencies, improving tracking precision by 20%.
- Developed a PID motor control algorithm utilizing optical encoder feedback and ADC signal processing, enhancing motor response time by 15%.

LED Frequency Filter

for Linear Systems, ELEE 6210

Athens, GA

August 2024 – December 2024

- Designed a frequency filter circuit to classify signals into predefined bands and visualize them via LEDs, achieving 98% signal classification accuracy.
- Iterated hardware design through oscilloscope analysis and voltage optimization to minimize signal attenuation and distortion.

Creation and Development of Websites

for a campus engineering student organization

Athens, GA

September 2022 – May 2024

- Spearheaded interactive WordPress and JavaScript website development, enhancing digital presence and increasing site traffic by 500+ monthly visitors.
- Coordinated with 10+ stakeholders to define requirements and apply UX principles, ensuring multi-device compatibility and seamless user experience.

Travel Itinerary Application

for Software Development, CSCI 1302

Athens, GA
December 2023

- Created JavaFX-based GUI application integrating Google Places and other RESTful APIs for dynamic data loading on hotels, attractions, and restaurants.
- Engineered efficient background threading to enable smooth UI responsiveness during data retrieval, increasing user satisfaction.

PROFESSIONAL EXPERIENCE

Capital One, Business Analyst Intern, McLean, VA

June 2025 – August 2025

- Spearheaded development of a business case for a Notifications Preferences Center for CreditWise, projected to streamline customer communication management for 60M+ users.
- Analyzed performance of CreditWise email campaigns, creating a valuation framework to quantify engagement and retention impact and propose a new email domain.
- Partnered with cross-functional teams to present actionable recommendations to senior leadership, driving alignment on future messaging strategy.
- Leveraged SQL, Python, Excel, and data visualization tools to evaluate KPIs, delivering insights that informed product roadmap decisions.

Great American Cookies & Marble Slab Creamery, Shift Leader, Dallas, GA

May 2022 – July 2022

- Managed daily operations and supervised a team of 5 employees, increasing shift efficiency by 20%.
- Processed 100+ customer orders daily with exemplary service, achieving top customer satisfaction ratings.
- Trained new staff, improving onboarding time by 40%.

HONORS & AWARDS

Extraordinary Engineer

February 2024

- Recognized by the College of Engineering for leadership, academic excellence, and service to the engineering community.

Presidential Scholar (2x)

December 2022, May 2023

- Awarded for maintaining a 4.0 GPA during consecutive semesters of full-time coursework.

Dire Needs Project Fund: \$1500 Project Grant

August 2023 – May 2024

- Selected through a competitive process to receive funding for a student-led project addressing a pressing engineering challenge.

CERTIFICATIONS

Emerging Engineering Leadership Development (EELD) Program

May 2026

- UGA College of Engineering Capstone Design program, developed in partnership with the J.W. Fanning Institute for Leadership Development, covering conflict resolution, stakeholder engagement, and cross-generational collaboration.

CITI Program: SBE Foundations, Human Subjects Research

September 2025

- Foundational training in the ethical and regulatory standards for human subjects research at the University of Georgia (Record ID 72070654; valid through September 2030).

LEADERSHIP & UNIVERSITY SERVICE

Tau Beta Pi Honor Society, Member

October 2024 – present

- Selected for academic distinction, leadership, and commitment to ethical engineering practice.
- Participate in professional development forums and community service events promoting excellence in STEM education.

Theta Tau Fraternity, Iota Epsilon Chapter, Brother

January 2024 – present

- Co-organize technical workshops, speaker panels, and community outreach efforts enhancing professional growth for 120+ members.
- Foster fraternity values of brotherhood, service, and lifelong learning through active participation in chapter initiatives.

UGA Engineering Student Society, *Vice President*

May 2025 – May 2026

- Lead a 100+ member chapter in strategic planning, aligning operations with national initiatives and regional directives.
- Coordinate logistics for national and regional conventions, managing travel, accommodations, and registration for 50+ members.
- Design and implement a centralized digital resource hub, providing internship pipelines, alumni contacts, and project archives to boost member success.

UGA Engineering Student Society, *Senator*

May 2024 – May 2025

- Represented chapter at regional and national levels, voting on legislation and advocating for student-centered initiatives.
- Led conference interest meetings, managing all travel logistics and budget allocations in coordination with the Treasurer.

UGA Engineering Student Society, *Telecommunications & Vice Public Relations Chair*

May 2023 – May 2024

- Increased chapter social media engagement by 40% through data-driven content strategies and visual design.
- Designed and launched a new chapter website to streamline communication, centralize resources, and showcase events.

TECHNICAL SKILLS

Programming: Assembly, C, C#, C++, HTML, Java, JavaFX, JavaScript, MATLAB, Python, R, SQL, TypeScript, Verilog

Software: Autodesk Fusion 360, Blender, CAD, Figma, Git, GitHub, Graphic Design, Klaviyo, Microsoft Suite, NASA F Prime, Next.js, Node.js, PlatformIO, React Native, Sanity CMS, Shopify Storefront API, Supabase, Tailwind CSS, Unity3D, Vercel, Virtual & Mixed Reality Development, Website Development, Wix, WordPress, Xilinx, Zephyr

AI & Machine Learning: Anthropic Claude API, LLM Application Development, AI Agents & Tool Use, Prompt Engineering, REST-based AI Integration, Structured (JSON) Outputs, Wit.ai Natural Language Understanding, Speech Synthesis & Text-to-Speech

Hardware and Embedded Systems: 2U CubeSat, Basys2 FPGA Boards, BLE / NimBLE, Cleanroom Microfabrication, ESP32, Geophones & Load Cells, Microfluidics, MQTT, PDMS Soft Lithography, Photolithography, Raspberry Pi Pico 2W, Raspberry Pi 4, Sensors, Signal Processing, STM32 Microcontrollers

Core Competencies: Business Case Development, Collaboration, Data Analysis, Critical Thinking, Human-Computer Interaction, Problem Solving, Product Strategy, Project Management, Real-Time Systems, Strategic Leadership, Technical Communication